

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	Oct 2 nd 2026
Team ID	6ab8c330c23107aaaa72e953
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	

Technical Architecture:

The project imports employee information from a spreadsheet into ServiceNow using Import Sets and Transform Maps. The spreadsheet contains Employee ID, Name, Email, Department, and Location. The data is first stored in an Import Set Table for temporary processing and validation. A Transform Map maps the source fields to the corresponding fields in the target Employee table. Employee ID is used as the unique identifier so existing employee records are updated and new employees are inserted .

Example: Employee information is imported from a spreadsheet into ServiceNow using Import Sets and Transform Maps. The Employee ID is used as a unique identifier to update existing employee records or insert new employee records.

Technical Architecture:

The following diagram shows the high-level architecture of the solution.



Guidelines:

1. Create a spreadsheet with the required employee details (Employee ID, Name, Email, Department, Location).
2. Upload the spreadsheet to ServiceNow using the Import Set.
3. Configure the Transform Map to map source fields to the Employee table fields.
4. Use Employee ID as the unique identifier to update existing records or insert new records.
5. Validate the data and check the Employee table for successful import.
6. Ensure proper error handling and logging for any failed records.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface through which the user uploads the employee spreadsheet and starts the data import process.	ServiceNow User Interface
2	Application Logic-1	Stores the uploaded spreadsheet data temporarily for processing and validation.	ServiceNow Import Sets
3	Application Logic-2	Maps the fields from the spreadsheet to the corresponding fields in the Employee table.	ServiceNow Transform Maps
4	Application Logic-3	Uses Employee ID as the unique identifier to update existing employee records or insert new employee records.	Transform Map with Coalesce
5	Database	Stores the imported employee information and the processed employee records.	ServiceNow Database
6	Cloud Database	Provides cloud-based storage for the employee records within the ServiceNow platform.	ServiceNow Cloud Platform

7	File Storage	Stores the spreadsheet file containing employee information such as Employee ID, Name, Email, Department, and Location.	Excel Spreadsheet / CSV File
8	External API-1	No external API is required for the employee data import process.	Not Applicable
9	External API-2	No external API is required for the employee data import process.	Not Applicable
10	Machine Learning Model	No machine learning model is required for importing and transforming employee data.	Not Applicable
11	Infrastructure (Server / Cloud)	The application is deployed and executed on the ServiceNow cloud platform.	ServiceNow Cloud Platform

Table-2:Application Characteristics:

S.N o	Characteristics	Description	Technology
1.	Open-Source Frameworks	No separate open-source framework is required for the project. The application uses the built-in features of the ServiceNow platform.	ServiceNow Platform
2.	Security Implementations	Access to employee data and import operations can be controlled using user roles and access controls.	ServiceNow Roles and ACLs
3.	Scalable Architecture	The application supports the processing of employee data through Import Sets and Transform Maps, allowing multiple records to be imported and processed efficiently.	ServiceNow Import Sets and Transform Maps
4.	Availability	The application is hosted on the ServiceNow cloud platform, allowing users to access the import functionality through the ServiceNow instance.	ServiceNow Cloud Platform
5.	Performance	Performance can be improved by processing employee data in batches and using Employee ID as the unique identifier for efficient record matching.	ServiceNow Import Sets and Transform Maps

References:

1. [ServiceNow Developer – Getting Started with Import Sets](#)
2. [ServiceNow Developer – Coalescing](#)
3. [ServiceNow – Importing and Transforming External Data](#)
4. [ServiceNow Developer – Create a Transform Map](#)